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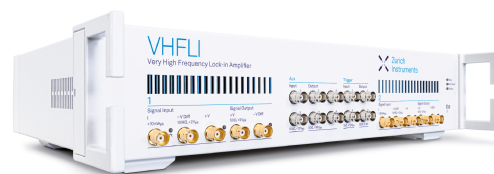
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SDE Model of SARS Disease in Hong Kong and Singapore with Parameter Stochasticity

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Abstract. A Stochastic Differential Equation (SDE) of SARS disease will be discussed in this article involving the randomness of transmission and recovery rate parameters. Intervention of medical mask and medical treatment will be included in the model in order to get a better understanding about a possibility of scenario to control the spread of SARS. Based on the infectious data from Singapore and Hong Kong, the parameters of the model will be approximated by the Least Square method. Numerical simulations of infected individual will be derived by the Euler-Maruyama method. Numerical simulations investigate some possible intervention scenarios : using medical mask, medical treatment, and both interventions.

Keywords: Stochastic differential equation, SARS, medical mask, medical treatment, Euler-Maruyama method

INTRODUCTION

Severe Acute Respiratory Syndrome (SARS) is one of a kind of respiratory disease. This disease caused by a Corona Virus (SARS-CoV). In General, corona virus infects animal and the virus can be transmitted to people. The main symptoms of SARS are high fever, headache, cough, difficulty breathing, and developing pneumonia. The first case is founded in Guangzhou, China in 2002. SARS spreads in some countries in Asia, Europe, and America. WHO received 8.098 cases and 774 died during the spread of SARS.

A mathematical model of SARS has been introduced by many authors, such as Ref. [1–4] with deterministic approaches, and [5, 6] with stochastic approaches. A SEIQR model in Ref. [1] suggests to do an early quarantine and to use a high quarantine rate effect to control SARS. A double epidemic model is used in Ref. [2] to see the probability that SARS is affected by unrelated viruses. Deterministic and stochastic SI in single area are compared in Ref. [5].

Based on Ref. [7–9], parameters that are considered in the epidemiological model have a stochastic behavior, for example is the probability of successful infection rate in SIR model. The stochasticity of these parameters mainly represent how the value of parameters might change depend on the environmental factors.

Related to the above description, a mathematical model of SARS disease spreading will be constructed in this article. The intervention of medical mask and medical treatment will be involved into the model due to suppress the number of infected people. The mathematical model construction of SDE model, where the probability of transmission and recovery rate are considered as stochastic parameters, will be given in the next section. In the third section, some numerical simulations will be performed using infectious data from Singapore and Hong Kong. The data will be used to test the model. In the last section, some conclusions will be given.

MATHEMATICAL MODEL CONSTRUCTION

We construct a model of SARS spreading with the assumption that individuals who have recovered from infection have no permanent immunity, so that they can be infected again soon after recovered from infection. The spread of

SARS disease can be modeled with the basic SIS model :

$$\frac{dS}{dt} = \mu N - \beta \frac{SI}{N} + \gamma_0 I - \mu S, \quad (1)$$

$$\frac{dI}{dt} = \beta \frac{SI}{N} - (\gamma_0 + \mu)I. \quad (2)$$

This model has two classes of population : S and I which are the number of susceptible and infected population at time t , respectively. We assume that the total population is constant, $S + I = N$, with μ is natural death rate, β is a transmission rate of SARS, while γ_0 is a natural recovery rate. The susceptible population may increases by the number of new born μN and number of recovered individual with rate of γ_0 . This population may decreases caused by death rate μ and infected with transmission rate of β . The transmission process is assumed to be perfectly random between individuals and described as a multiplication between number of infected individual (I) and the proportion of susceptible human ($\frac{S}{N}$). The susceptible individual who gets infected move out to the infected population (I), which will increase number of infected population. However, the infected population decrease because of recovery term which transfer infected human into the susceptible population (S) and the death rate μ .

In this article, we will improved above model 1 with introducing interventions by using medical mask and medical treatment. The proportion of individual who using medical mask and medical treatment of infected class are u_1 and u_2 , respectively. Infected people who use a medical mask has a positive effectiveness p and infected people who undergo with medical treatment, will increase their recovery rate from γ_0 into $\gamma_0 + \gamma_1$. The differential equations become :

$$\frac{dS}{dt} = \mu N - \beta \frac{SI}{N} (u_1(p-1) + 1) + (\gamma_0 + \gamma_1 u_2)I - \mu S, \quad (3)$$

$$\frac{dI}{dt} = \beta \frac{SI}{N} (u_1(p-1) + 1) - (\gamma_0 + \gamma_1 u_2 + \mu)I. \quad (4)$$

The deterministic model above does not count uncertainty factors. One of the uncertainty factors is probability of successful infection from infected to susceptible population [9]. Others factor that can be categorized as an environment stochasticity are the hygiene, and lifestyle of human. With this reasoning, transforming above model 3 into a stochastic differential equation model will accommodate these stochasticity factors.

This modern, stochastic differential equation model uses the perturbation parameters to represent the effect of these random factors on the disease spread. Regarding to this purpose, noise factors are added to the rate of transmission parameter β since there is a probability of success infection between infected and susceptible individuals and also to the natural recovery rate parameter γ_0 because there is an environment impact to the recovery. Therefore, we change the transmission rate βdt and the recovery rate $\gamma_0 dt$ at time $[t, t + dt]$ with $\tilde{\beta} dt$ and $\tilde{\gamma}_0 dt$, where

$$\tilde{\beta} dt = \beta dt + \sigma_1 dW_1(t)$$

$$\tilde{\gamma}_0 dt = \gamma_0 dt + \sigma_2 dW_2(t),$$

with σ_i for $i = 1, 2$ are the noise coefficients while $dW(t)$ is the increment of Brownian Motion. Therefore, the model 3 become :

$$dS = \left(\mu N - \beta \frac{SI}{N} (u_1(p-1) + 1) + (\gamma_0 + \gamma_1 u_2)I - \mu S \right) dt - \sigma_1 \frac{SI}{N} (u_1(p-1) + 1) dW_1(t) + \sigma_2 I dW_2(t), \quad (5)$$

$$dI = \left(\beta \frac{SI}{N} (u_1(p-1) + 1) - (\gamma_0 + \gamma_1 u_2 + \mu)I \right) dt + \sigma_1 \frac{SI}{N} (u_1(p-1) + 1) dW_1(t) - \sigma_2 I dW_2(t). \quad (6)$$

NUMERICAL SIMULATION

In this section, the dynamics of deterministic model and stochastic model with or without interventions are investigated. Since the spread of SARS in the country can be seen from the dynamic of infected population, in this simulation we will focus only on $dI(t)$ in the Equation 2b and 3b. In the simulation, we use the daily SARS cases data on Singapore and Hong Kong on 17 March 2003 to 23 July 2003 [10].

Firstly, we would like to approximate the daily incidence data by choosing a best parameters value in model 3 such that the error between the dynamic of the model and daily incidence data could be minimum as possible. The death

rate parameter μ and total population N in Singapore and Hong Kong [11] are in the following Table 1. However, the transmission rate β and the natural recovery rate γ_0 are estimated by least square fitting method.

We estimate the transmission rate β and the natural recovery rate γ_0 by using least-square fitting method. Input the parameter values in Table 1 in the deterministic model 1. Then, set the initial values for Singapore as $\beta = 0,68/\text{day}$, $\gamma_0 = 0,125$ and for Hong Kong as $\beta = 0,75/\text{day}$, $\gamma_0 = 0,125$ [12]. The parameters are evaluated by Nelder-Mead algorithm with tolerance 10^{-4} . The Nelder-Mead algorithm for minimizing the function of n variables with $n + 1$ simplex. The approximated parameters for Singapore are $\beta = 0,109/\text{day}$, $\gamma_0 = 0,213/\text{day}$ and for Hong Kong are $\beta = 0,171/\text{day}$, $\gamma_0 = 0,210/\text{day}$.

Secondly, using the parameters in Table 1 and approximated parameter before, we will mimic the behavior of daily incidence of SARS data on Singapore and Hong Kong with the Euler-Maruyama Method using system 5 where $u_1 = u_2 = 0$. We take 100.2^{10} as the discretized Brownian paths and 2^8 as the step-size of Euler-Maruyama and compute over 1000 paths. The numerical simulations based on Singapore and Hong Kong data can be seen in Fig. 1, 2 and 3 for various value of σ .

We will see the effect of coefficient noise over the dynamics of SARS spreading without intervention in Equation 5 by set $u_1 = u_2 = 0$. Green, blue, and red curves are the mean of 1000 simulations for each value of σ . In Fig. 1, we give various value noise coefficients of the transmission rate ($\sigma_1 = 0, 0.2, 0.4$) while the noise coefficient of the

TABLE 1. Death rate parameter μ and total population N in Singapore and Hong Kong [11]

	Singapura	Hong Kong
N	4.115.000 people	6.731.000 people
μ	$\frac{1}{79 \times 365 \text{ day}}$	$\frac{1}{81 \times 365 \text{ day}}$
$I(0)$	20 people	95 people

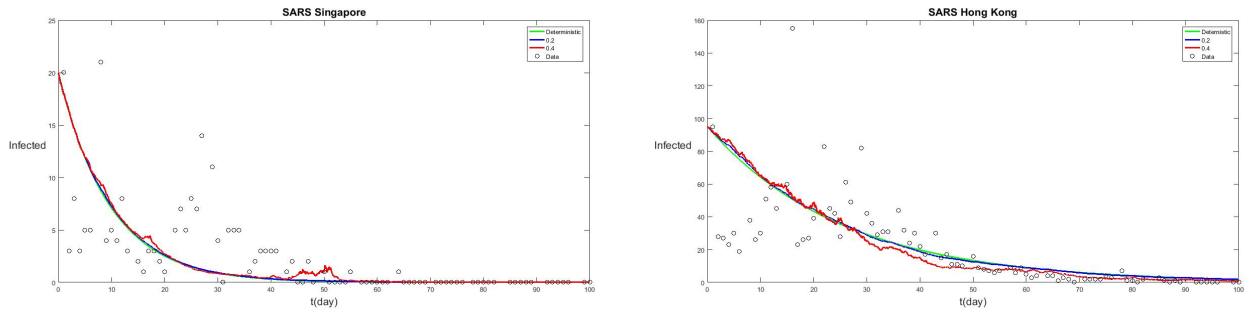


FIGURE 1. Dynamic of infected human for Singapore (left) and Hong Kong (right) SARS data with various value of σ_1 , while $\sigma_2 = 0$.

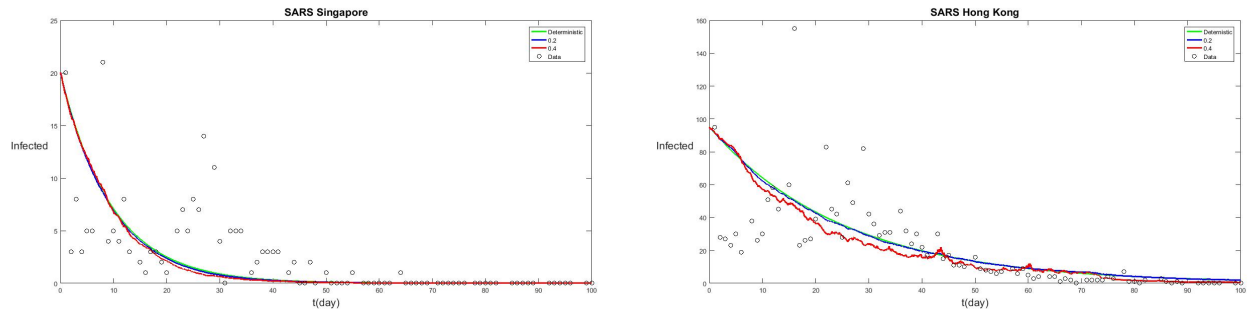


FIGURE 2. Dynamic of infected human for Singapore (left) and Hong Kong (right) SARS data with various value of σ_2 , while $\sigma_1 = 0$.

recovery rate is not counted ($\sigma_2 = 0$). In Fig. 2, we do not include the coefficient of transmission rate ($\sigma_1 = 0$) but have various coefficients noise of recovery rate ($\sigma_2 = 0, 0.2, 0.4$). Please note that the choosing of these σ values are based on qualitatively determine how random is the environment. Larger value of σ describe more stochastic is the environment.

By comparing Fig. 1 and 2, it can be seen the noise coefficient do not give the significant differences in Singapore but in Hong Kong quite obvious when $\sigma_1 = 0.4$ and $\sigma_2 = 0.4$. In Singapore, the fluctuation can be seen slightly when the noise coefficient of transmission rate (σ_1) is include. So that, the stochastic factor for infection probability between infected and susceptible individual has shown a more significant effect. Even though the noise coefficient is large, the stochastic model can only present the data nicely in $t > 60$. In Hong Kong, the noise coefficient of recovery rate (σ_2) that denote the stochastic factor for recovery rate has shown a more significant effected. The red curve can present the Hong Kong data in $40 < t < 60$ and $t > 80$ nicely.

To investigate the effect for both noise coefficient, we set the noise coefficient for same value ($\sigma_1 = \sigma_2 = 0, 0.2, 0.4$) in Fig. 3. For dynamics when $\sigma_1 = \sigma_2 = 0.4$, infected simulation in Hong Kong present the fluctuate more clearly than infected simulation in Singapore. Include the both noise make the fluctuation is more clear than only one noise.

Next, we will use the same method to show the effect of intervention regarding Equation 5. We have three scenarios when the infected people is using a mask, getting medical treatment, and both. We use the same values from Table 1, the estimated parameters β and γ_0 , $\sigma_1 = 0.2$, and $\sigma_2 = 0.2$. We present the mean of 1000 simulations for each scenario. Figure 4 and 5 shows the effect of various value of infected people who using a medical masks ($u_1 = 0, 0.3, 0.6, 0.9$ with the effect $p = 0.3$) and medical treatment ($u_2 = 0, 0.3, 0.6, 0.9$ with medical recovery rate $\gamma_1 = 1/14$ per day), respectively. Figure 6 shows the effect of medical mask and medical treatment when implemented together to infected people ($u_1 = 0, 0.3, 0.6, 0.9$ with $p = 0.3$ and $\gamma_1 = 1/14$ per day). In general, the proportion of infected population who using a medical mask only, getting a medical treatment only, or both scenarios will accelerate the disappearance of the disease. In Fig. 4, 5, and 6, the effect of interventions are similar on Singapore and Hong Kong data. The infected people disappear in $30 < t < 40$ in Singapore and in $50 < t < 60$ in Hong Kong when 90% of

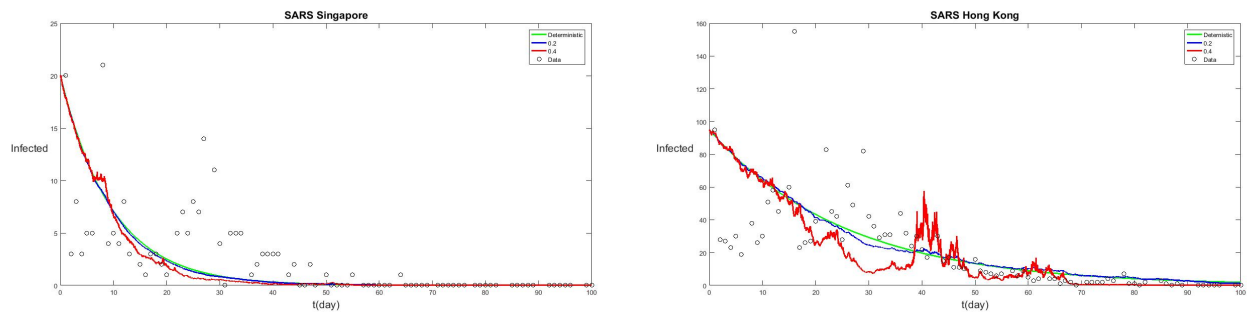


FIGURE 3. Dynamic of infected human for Singapore (left) and Hong Kong (right) SARS data with various value of σ_1 and σ_2 .

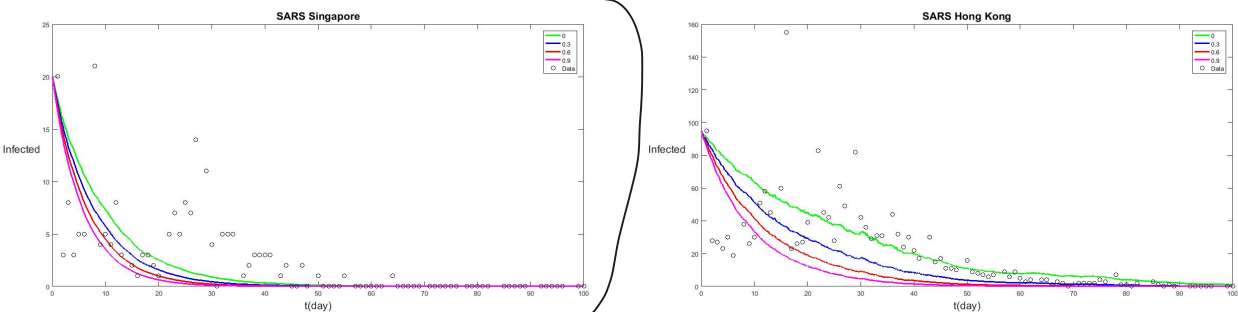


FIGURE 4. Scenario 1 : use of a medical mask with values are $u_1 = 0, 0.3, 0.6, 0.9$, $p = 0.3$, and $u_2 = 0$

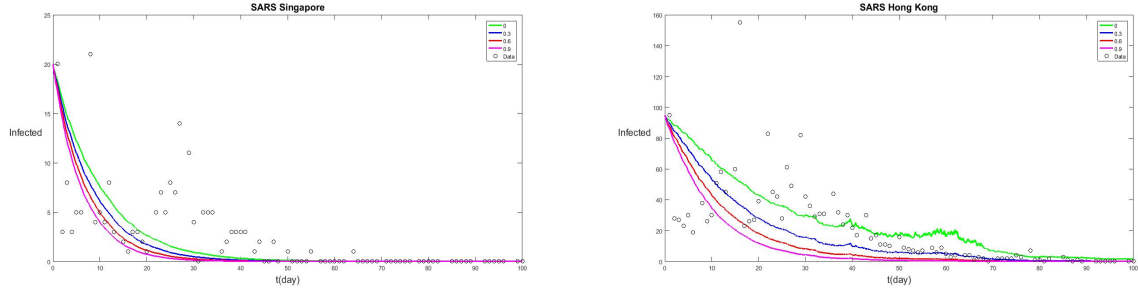


FIGURE 5. Scenario 2 : medical treatment with values are $u_1 = 0, u_2 = 0, 0.3, 0.6, 0.9$, and $\gamma_1 = 1/14$ per day.

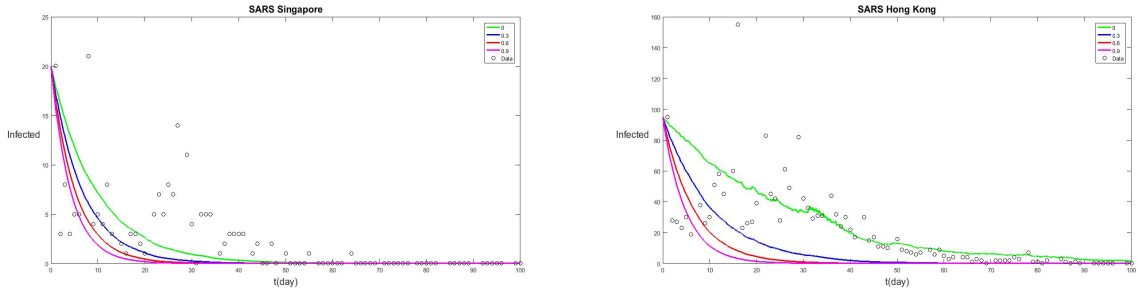


FIGURE 6. Scenario 3 : use a mask and medical treatment with values are $u_1 = u_2 = 0, 0.3, 0.6, 0.9, p = 0.3$ and $\gamma_1 = 1/14$ per day.

infected people is using a medical mask or getting a medical treatment. In the other hand, it can be seen in Fig. 6 that the infected people disappear in $20 < t < 30$ for both countries. Therefore, intervention of using medical mask and medical treatment is success to reduce the spread of SARS significantly.

CONCLUSION

A mathematical model of SARS spreading in Hong Kong and Singapore with an SI model is introduced in this article. The SDE model is constructed from the deterministic model which considering the stochasticity of infection probability and natural recovery rate into the model. Intervention of medical mask and medical treatment is included into the model to see how these interventions could reduce daily incidence in Hong Kong and Singapore. From numerical simulation to finding the best parameter value to approximate the daily incidence data in both cities, it can be seen that infection probability in Hong Kong is larger than in Singapore, while the natural recovery parameters in both cities is only slightly different.

From SDE simulation, we find that noise coefficient which describe the stochasticity of infection probability is giving much more effect in fluctuation of infected people dynamics rather than with stochasticity of natural recovery rate.

Although the model in this article is not pretty well to capture whole daily incidence data in Singapore and Hong Kong, the model still could to improved with considering a more realistic fact about SARS in the field, such as the characteristic of the virus.

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